PHYSICS 4

ACADEMIC

Grade 12

Curriculum Guide

PHYSICS 4
Prerequisites: Physics 1, Physics 2, Physics 3

Course Description:

Physics 4 builds on the knowledge and skills developed in Physics 3. The course focuses on the behavior and applications of waves and light, as well as the structure of the atom and the principles of nuclear physics. The learners explore the properties of mechanical and electromagnetic waves and how these waves interact with matter through phenomena such as reflection, refraction, diffraction, interference, polarization, and the Doppler effect.

The course also introduces concepts in atomic and nuclear physics, including atomic structure, radioactivity, nuclear reactions, and the interaction of radiation with matter. Through problem solving, analysis of real-world applications, and investigation of modern technologies, the learners develop a deeper understanding of how wave and nuclear phenomena are applied in fields such as communication, medicine, energy production, and industry.

Elective: Academic
Time Allotment: 80 hours for one term

CONTENT	**CONTENT STANDARDS** *The learners demonstrate understanding that…*	**LEARNING COMPETENCIES** *The learners…*
Waves and Optics		
1. Mechanical Waves 2. Electromagnetic waves 3. Applications of Sound and Light	1. the nature and behavior of sound and light waves have diverse applications in various fields such as acoustics, communication technology,	1. differentiate mechanical waves and electromagnetic waves based on nature of propagation and examples;
		2. characterize waves based their parts (crest, trough, compression, rarefaction) and properties (wavelength, amplitude, frequency, period);

CONTENT	**CONTENT STANDARDS** *The learners demonstrate understanding that…*	**LEARNING COMPETENCIES** *The learners…*
	medicine, optics, and entertainment.	3. describe light as an electromagnetic wave and relate its properties, such as wavelength and frequency, to its behavior when interacting with different media;
		4. analyze how waves interact with different media and obstacles through reflection, refraction, diffraction, and interference;
		5. determine the type, orientation, position, and magnification of images formed by spherical mirrors and thin lenses using ray diagrams and mathematical equations;
		6. describe how the properties of sound, such as pitch, loudness, and quality (timbre), are related to wave characteristics;
		7. use the properties of sound to explain real-world scenarios such as echoes, soundproofing, acoustics, and noise cancellation;
		8. explain how the properties of light are applied in modern technologies such as fiber-optic communication, laser-based devices, optical sensors, and medical imaging systems; and

CONTENT	CONTENT STANDARDS *The learners demonstrate understanding that…*	LEARNING COMPETENCIES *The learners…*
		9. explain frequency shifts in sound and light caused by relative motion between sources and observers using the Doppler effect in contexts such as moving vehicles, stellar motion, and atmospheric monitoring.
Performance Standard *The learners* can explain the properties and behaviors of mechanical and electromagnetic waves through theoretical analysis and contextual problem solving, as well as wave phenomena and their applications in sound, light, communication, and imaging technologies.		
Suggested Performance Task Create a simulation model of waves and acoustics using everyday materials. The model should accurately show the nature and behavior of either sound or light waves.		
Atomic and Nuclear Physics		
4. Atomic Structure and Nuclear Composition 5. Nuclear Forces, Stability, and Radioactivity	2. competing forces within the nucleus produce stable and unstable energy configurations; 3. unstable nuclei become more stable through nuclear transformations;	10. explain the structure of the atom in terms of the nucleus, electrons, protons, and neutrons, and the quark composition of nucleons;
		11. describe how protons and neutrons are composed of quarks and relate this to the study of particle physics and particle accelerators;
		12. explain the concept of binding energy, strong nuclear force and its role in nuclear stability, and occurrence of radioactivity;

CONTENT	CONTENT STANDARDS *The learners demonstrate understanding that…*	LEARNING COMPETENCIES *The learners…*
		13. analyze properties and use of specific elements and their isotopes using the periodic table of elements and chart of nuclides;
6. Radioactive Decay and Half-life 7. Radiation Types, Interaction with Matter, and Applications	4. ionizing radiation has both beneficial and harmful effects, making radiation safety and protection essential;	14. apply dosimetric quantities and units to distinguish ionizing from nonionizing radiation and assess radiation safety;
		15. analyze the processes of radioactive decay, including alpha, beta, and gamma decay;
		16. calculate the half-life of radioactive substances and binding energy and mass energy in nuclear reactions;
8. Nuclear Reactions and Technologies	5. energy stored within mass of objects can be converted into other forms of energy and vice versa; and 6. radiation is a form of energy transmitted by rays or beams of particles.	17. compare nuclear fission and fusion in terms of their principles and implications for energy production, safety, and radioactive waste management;
		18. describe the interaction of radiation with matter and its applications in medicine, agriculture, industries, and environmental monitoring; and
		19. use secondary sources to explain the principles of particle accelerators, reactors, and cyclotrons and their uses.

CONTENT	CONTENT STANDARDS *The learners demonstrate understanding that…*	LEARNING COMPETENCIES *The learners…*
Performance Standard *The learners* can explain the principles of atomic and nuclear physics through theoretical analysis and contextual problem solving and can recognize their applications and implications for nuclear energy and radiation safety.		
Suggested Performance Task Write a position paper on the practical application and safety of radiation to a select field.		